Benjamin Johnston

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Technical Skills

Languages: C, C++, C#, D, x86 Assembly, Python
Systems & RE: Reverse Engineering, ARM64/x86 Disassembly, Firmware Analysis, PE Format, CoreCLR Internals, FFI/Interop, Template Metaprogramming
Security: Vulnerability Research, Hook Detection, Windows Syscalls, Shellcode Generation, HMAC, CVE Analysis
Performance: SIMD/AVX-512, SSE, BMI2, AArch64 NEON, Custom Allocators, Cache-friendly Design
Infrastructure: Linux, Git, CMake, Docker, CI/CD

Experience

Software Engineer Jul 2021 – Present

Independent client engagements

- Reverse-engineered the VEX Robotics V5 private firmware API from ARM64 disassembly, documenting 8,500+ lines of C/C++ headers across 39 files with 481 firmware memory offsets; cited by the Rust vex-sdk crate.
- Designed and maintain 10+ open-source libraries in C, C#, C++, and D (700+ commits), spanning runtime introspection (Cassowary: 15 CoreCLR structures), Windows syscall research (UndeadInterop: 8 hook detection classifications), and a compile-time multi-ISA assembler (Multidasm: 651 AMD64 instructions).

Automation Engineer Nov 2025 – Present

Handshake

- Built AutoHandshake!, a cross-platform (Linux, macOS, Windows) Selenium automation suite with unattended execution and UI-stall recovery, eliminating 95% of manual effort across 400+ runtime hours.

AI Evaluation Engineer Dec 2025 – Jun 2026

Snorkel AI

- Built Docker-based reproducible evaluation sandboxes with strict schema constraints, catching defects pre-release and enforcing training-set integrity before pipeline ingestion.

Selected Projects

VEXAPI | *Firmware Reverse Engineering, C/C++* | github.com/cetio/VEXAPI

- Reverse-engineered the VEX Robotics V5 private firmware API from ARM64 disassembly of VEXos using Ghidra, documenting 8,500+ lines of C/C++ headers across 39 files with 481 firmware memory offsets.
- Produced 34 public API headers covering 20+ device types and 5 private headers with jump table and boot sequence signatures; cited by the Rust vex-sdk crate.

UndeadInterop | *Windows Syscall Research, C#* | github.com/cetio/UndeadInterop

- Resolved Windows syscall IDs from PE export tables and invoked them directly via runtime-generated x86_64 shellcode stubs, bypassing ntdll usermode hooks.
- Implemented 8 hook detection classifications (inline jumps/calls, stack pivots, debug interrupts, forwarded exports) via Iced x86 decoder, with a complete NTSTATUS enum (339 values).

Multidasm | *Multi-ISA Assembler, D* | github.com/cetio/multidasm

- Built a compile-time assembler targeting x86, x86_64, CIL, and MIPS, with 651 AMD64 instruction template instantiations (79% coverage) and 226 CIL opcodes (100% coverage).
- Implemented SIMD instruction encoding (MMX through AVX2/FMA) with label resolution, patching, and block-based API with executable memory emission.

Cassowary | *.NET Runtime Introspection, C#* | github.com/cetio/Cassowary

- Authored a managed C# library for .NET runtime introspection, modeling 15 CLR internal structures (MethodTable, EEClass, MethodDesc, FieldDesc) and 15 type signature models.
- Implemented bi-directional C++ interop with unsafe pointer-based access and Reflection.Emit delegate construction; referenced by Novus, one of the largest .NET reflection libraries.

Education

Columbus State Community College Expected Dec 2026

AAS, Software Development